```
BITS 32
GLOBAL _start

%macro PRINT 2
   mov ebx, 1
   mov ecx, %1
   mov edx, %2
   call write_all
%endmacro

%macro SAVE 2
   mov ebx, [receipt_fd]
   mov ecx, %1
   mov edx, %2
   call write_all
%endmacro

SECTION .data
   filename db "receipt.txt", 0

   menu db 10
       db "+-----------------------------------------------------+", 10
       db "|                GEORGETTE EATERY                     |", 10
       db "+-----------------------------------------------------+", 10
       db "| A. Rice: Plain Rice               Php  5.00       |", 10
       db "| B. Rice: Java Rice                Php 10.00       |", 10
       db "| C. Ulam 1: Fish                   Php 35.00       |", 10
       db "| D. Ulam 2: Pork                   Php 50.00       |", 10
       db "| E. Ulam 3: Vegetables             Php 20.00       |", 10
       db "| F. Dessert: Saging con yelo       Php 60.00       |", 10
       db "+-----------------------------------------------------+", 10
   menu_len equ $ - menu

   order_prompt db "Enter your order [A-F]: "
   order_prompt_len equ $ - order_prompt

   ordered_msg db "You ordered "
   ordered_msg_len equ $ - ordered_msg

   again_prompt db "Do you want to order again? [Y/N]: "
   again_prompt_len equ $ - again_prompt

   invalid_order db "Invalid order. Please enter A to F.", 10
   invalid_order_len equ $ - invalid_order
```

```
invalid_again db "Invalid answer. Please enter Y or N.", 10
invalid_again_len equ $ - invalid_again

thanks db 10, "Thank you! Please come again!", 10, 10
thanks_len equ $ - thanks

file_error db "Error: Unable to read or write the receipt.", 10
file_error_len equ $ - file_error

input_error db "Error: Unable to read input.", 10
input_error_len equ $ - input_error

receipt_header db "======================================", 10
               db "          GEORGETTE EATERY", 10
               db "               RECEIPT", 10
               db "======================================", 10
receipt_header_len equ $ - receipt_header

receipt_footer db "======================================", 10
               db "Thank you! Please come again!", 10
receipt_footer_len equ $ - receipt_footer

plain_rice db "Plain Rice: Php 5.00", 10
plain_rice_len equ $ - plain_rice

java_rice db "Java Rice: Php 10.00", 10
java_rice_len equ $ - java_rice

fish db "Fish: Php 35.00", 10
fish_len equ $ - fish

pork db "Pork: Php 50.00", 10
pork_len equ $ - pork

vegetables db "Vegetables: Php 20.00", 10
vegetables_len equ $ - vegetables

dessert db "Saging con yelo: Php 60.00", 10
dessert_len equ $ - dessert

items:
    dd plain_rice, plain_rice_len
    dd java_rice, java_rice_len
```

```
        dd fish, fish_len
        dd pork, pork_len
        dd vegetables, vegetables_len
        dd dessert, dessert_len

SECTION .bss
    receipt_fd resd 1
    key resb 1
    receipt_buffer resb 512

SECTION .text

_start:
    call create_receipt

order_loop:
    call display_menu

get_order:
    PRINT order_prompt, order_prompt_len
    call read_choice

    cmp eax, -1
    je finish_ordering

    sub eax, 'A'
    cmp eax, 5
    ja bad_order

    mov esi, [items + eax * 8]
    mov edi, [items + eax * 8 + 4]

    call display_order
    call save_order

ask_again:
    PRINT again_prompt, again_prompt_len
    call read_choice

    cmp eax, 'Y'
    je order_loop

    cmp eax, 'N'
    je finish_ordering
```

```
    cmp eax, -1
    je finish_ordering

    PRINT invalid_again, invalid_again_len
    jmp ask_again

bad_order:
    PRINT invalid_order, invalid_order_len
    jmp get_order

finish_ordering:
    SAVE receipt_footer, receipt_footer_len

    mov eax, 6
    mov ebx, [receipt_fd]
    int 0x80
    test eax, eax
    js fatal_file

    PRINT thanks, thanks_len
    call display_receipt

    mov eax, 1
    xor ebx, ebx
    int 0x80

create_receipt:
    mov eax, 8
    mov ebx, filename
    mov ecx, 0o644
    int 0x80

    test eax, eax
    js fatal_file

    mov [receipt_fd], eax
    SAVE receipt_header, receipt_header_len
    ret

display_menu:
    PRINT menu, menu_len
    ret
```

```
display_order:
    PRINT ordered_msg, ordered_msg_len
    PRINT esi, edi
    ret

save_order:
    SAVE esi, edi
    ret

read_choice:
    push ebx
    push ecx
    push edx
    push esi
    push edi

    xor esi, esi
    xor edi, edi

.read:
    mov eax, 3
    xor ebx, ebx
    mov ecx, key
    mov edx, 1
    int 0x80

    test eax, eax
    js fatal_input
    jz .eof

    movzx eax, byte [key]

    cmp al, 10
    je .line_end

    cmp al, 13
    je .read

    test esi, esi
    jnz .extra_character

    mov edi, eax
    mov esi, 1
    jmp .read
```

```
.extra_character:
    mov esi, 2
    jmp .read

.eof:
    test esi, esi
    jnz .line_end

    mov eax, -1
    jmp .done

.line_end:
    xor eax, eax
    cmp esi, 1
    jne .done

    mov eax, edi
    cmp al, 'a'
    jb .done
    cmp al, 'z'
    ja .done
    sub al, 32

.done:
    pop edi
    pop esi
    pop edx
    pop ecx
    pop ebx
    ret

write_all:
    pushad

.write:
    test edx, edx
    jz .done

    mov eax, 4
    int 0x80

    test eax, eax
    jle fatal_file
```

```
    add ecx, eax
    sub edx, eax
    jmp .write

.done:
    popad
    ret

display_receipt:
    mov eax, 5
    mov ebx, filename
    xor ecx, ecx
    int 0x80

    test eax, eax
    js fatal_file

    mov [receipt_fd], eax

.read:
    mov eax, 3
    mov ebx, [receipt_fd]
    mov ecx, receipt_buffer
    mov edx, 512
    int 0x80

    test eax, eax
    js fatal_file
    jz .close

    mov edx, eax
    mov ebx, 1
    mov ecx, receipt_buffer
    call write_all
    jmp .read

.close:
    mov eax, 6
    mov ebx, [receipt_fd]
    int 0x80

    test eax, eax
    js fatal_file
```

```
    ret

fatal_input:
    mov ecx, input_error
    mov edx, input_error_len
    jmp fatal_exit

fatal_file:
    mov ecx, file_error
    mov edx, file_error_len

fatal_exit:
    mov eax, 4
    mov ebx, 2
    int 0x80

    mov eax, 1
    mov ebx, 1
    int 0x80
```

```
root@1c666e8c76a2:/workspaces/hjtnasroot@1c666e8c76a2:/workspaces/hjtnasm# n-f elf32 georgette.asm -o george
asm -f elf32 georgette.asm -o georgette.o elf_i386 georgette.o -o george
ld -m elf_i386 georgette.o -o georgettegeorgette
./georgette

+--------------------------------------------------+
|                GEORGETTE EATERY                  |
+--------------------------------------------------+
| A. Rice: Plain Rice               Php  5.00      |
| B. Rice: Java Rice                Php 10.00      |
| C. Ulam 1: Fish                   Php 35.00      |
| D. Ulam 2: Pork                   Php 50.00      |
| E. Ulam 3: Vegetables             Php 20.00      |
| F. Dessert: Saging con yelo       Php 60.00      |
+--------------------------------------------------+
Enter your order [A-F]: A
You ordered Plain Rice: Php 5.00
Do you want to order again? [Y/N]: Y

+--------------------------------------------------+
|                GEORGETTE EATERY                  |
+--------------------------------------------------+
| A. Rice: Plain Rice               Php  5.00      |
| B. Rice: Java Rice                Php 10.00      |
| C. Ulam 1: Fish                   Php 35.00      |
| D. Ulam 2: Pork                   Php 50.00      |
| E. Ulam 3: Vegetables             Php 20.00      |
| F. Dessert: Saging con yelo       Php 60.00      |
+--------------------------------------------------+
Enter your order [A-F]: B
You ordered Java Rice: Php 10.00
Do you want to order again? [Y/N]: Y

+--------------------------------------------------+
|                GEORGETTE EATERY                  |
+--------------------------------------------------+
| A. Rice: Plain Rice               Php  5.00      |
| B. Rice: Java Rice                Php 10.00      |
| C. Ulam 1: Fish                   Php 35.00      |
| D. Ulam 2: Pork                   Php 50.00      |
| E. Ulam 3: Vegetables             Php 20.00      |
| F. Dessert: Saging con yelo       Php 60.00      |
```

```
+-------------------------------------------------+
|                GEORGETTE EATERY                 |
+-------------------------------------------------+
| A. Rice: Plain Rice                 Php  5.00   |
| B. Rice: Java Rice                  Php 10.00   |
| C. Ulam 1: Fish                     Php 35.00   |
| D. Ulam 2: Pork                     Php 50.00   |
| E. Ulam 3: Vegetables               Php 20.00   |
| F. Dessert: Saging con yelo         Php 60.00   |
+-------------------------------------------------+
Enter your order [A-F]: C
You ordered Fish: Php 35.00
Do you want to order again? [Y/N]: Y

+-------------------------------------------------+
|                GEORGETTE EATERY                 |
+-------------------------------------------------+
| A. Rice: Plain Rice                 Php  5.00   |
| B. Rice: Java Rice                  Php 10.00   |
| C. Ulam 1: Fish                     Php 35.00   |
| D. Ulam 2: Pork                     Php 50.00   |
| E. Ulam 3: Vegetables               Php 20.00   |
| F. Dessert: Saging con yelo         Php 60.00   |
+-------------------------------------------------+
Enter your order [A-F]: D
You ordered Pork: Php 50.00
Do you want to order again? [Y/N]: Y

+-------------------------------------------------+
|                GEORGETTE EATERY                 |
+-------------------------------------------------+
| A. Rice: Plain Rice                 Php  5.00   |
| B. Rice: Java Rice                  Php 10.00   |
| C. Ulam 1: Fish                     Php 35.00   |
| D. Ulam 2: Pork                     Php 50.00   |
| E. Ulam 3: Vegetables               Php 20.00   |
| F. Dessert: Saging con yelo         Php 60.00   |
+-------------------------------------------------+
Enter your order [A-F]: E
You ordered Vegetables: Php 20.00
Do you want to order again? [Y/N]: Y
```

```
+-------------------------------------------------+
|                GEORGETTE EATERY                 |
+-------------------------------------------------+
| A. Rice: Plain Rice                Php  5.00    |
| B. Rice: Java Rice                 Php 10.00    |
| C. Ulam 1: Fish                    Php 35.00    |
| D. Ulam 2: Pork                    Php 50.00    |
| E. Ulam 3: Vegetables              Php 20.00    |
| F. Dessert: Saging con yelo        Php 60.00    |
+-------------------------------------------------+
Enter your order [A-F]: F
You ordered Saging con yelo: Php 60.00
Do you want to order again? [Y/N]: N

Thank you! Please come again!


====================================
          GEORGETTE EATERY
                RECEIPT
====================================
Plain Rice: Php 5.00
Java Rice: Php 10.00
Fish: Php 35.00
Pork: Php 50.00
Vegetables: Php 20.00
Saging con yelo: Php 60.00
====================================
Thank you! Please come again!
root@1c666e8c76a2:/workspaces/hjtnasm#
```